



COMPUTER SCIENCE & ENGINEERING
Module 1: Introduction

</> Day_01.java

Java Programming

Basics for CSE Students

"Write once, run anywhere" JVM

TODAY'S LEARNING PATH

i What is Java & Why use it?

>- Hello World & Syntax

% How Code Runs (JVM)

Variables & Data Types

INTRODUCTION

What is Java?

Core Concepts

</> The Language

General Purpose & OOP

A robust, object-oriented programming language designed to build everything from small apps to large enterprise systems.

Platform Independent

"Write Once, Run Anywhere." Thanks to the JVM, the same code runs on Windows, Mac, or Linux without changes.

Versatile Usage

Powers Android apps, backend server systems, web applications, and desktop software worldwide.

Why for CSE Students?

Strong Foundation

Learning Java makes picking up C#, Kotlin, Scala, and C++ much easier due to similar syntax and concepts.

Industry Standard

A top requirement for big tech companies (Google, Amazon, etc.) and a favorite for coding interviews.

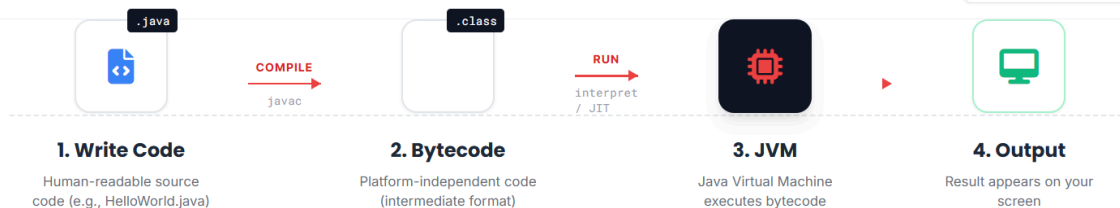
Mature Ecosystem

Huge community support, excellent documentation, and powerful tools (IDEs like IntelliJ, Eclipse) help you learn faster.

PROCESS OVERVIEW

How Java Code Runs

Execution Flow



Why this matters: "Write Once, Run Anywhere"

The magic happens at the **Bytecode** stage. You don't rewrite code for Windows, Mac, or Linux. You write it once, compile it into bytecode, and the JVM specific to that operating system runs it perfectly.

FIRST STEPS

Your First Program

> Hello, World!

```
1 public class HelloWorld {
2     public static void main(String[] args) {
3         System.out.println("Hello, World!");
4     }
5 }
```

Class Definition

1 The class name `HelloWorld` must strictly match the filename `HelloWorld.java`.

Main Method

2 `public static void main` is the entry point. Java looks for this exact method to start your program.

Print Statement

3 `System.out.println` prints the text inside the quotes to the console and moves to a new line.

THE RULES OF WRITING

Basic Java Syntax

≡ Key Requirements



Every Statement Ends with a Semicolon

Just like a period in English, a semicolon tells the compiler the instruction is complete.

```
System.out.println("Hello"); // Correct
```



Java is Case-Sensitive

Capitalization matters! `System` is not the same as `system`.

✓ main

✗ Main

(for method)



Curly Braces Define Blocks

They group code for classes, methods, and loops. Always keep them paired.

```
public class Demo {
    // code block starts here
}
```

Pro Tip: Indentation

Always use proper indentation inside braces (usually 4 spaces or 1 tab). It keeps your code readable and professional.

```
Class {
```



Avoid Compile Errors

Following these syntax rules prevents the dreaded "compile-time errors" before you even run your code.

NOTES FOR HUMANS

Comments in Java

Code Documentation



Ignored by the Compiler

Comments are notes for you and other developers. The Java compiler completely skips them, so they don't affect your program's performance or output.



Single-Line Comment

Use two forward slashes. Everything after them on the same line is ignored.

```
int age = 18; // This is a single-line comment
```



Multi-Line Comment

Starts with `/*` and ends with `*/`. Great for longer explanations spanning paragraphs.

```
/*
  This is a multi-line comment.
  It can span multiple lines.
*/
```

Best Practice

Start using comments from Day 1 to build good habits.

✗ Don't do this:
// Sets x to 10
x = 10;

```
// TODO: Fix this
/* Deprecated */
// Author: Admin
```



Clean Code

Comments explain the "Why", the code explains the "How".

CORE FUNDAMENTALS

Variables & Data Types

Storage & Formats

What is a Variable?

Think of a variable as a **named container** in your computer's memory where you store data values to use later.

COMMON DATA TYPES

int

Integer (Whole Numbers)

e.g., 10, -5, 2026

dbl

Double (Decimals)

e.g., 3.14, 99.99, -0.5

chr

Char (Single Character)

e.g., 'A', 'z', '\$' (Uses single quotes)

bool

Boolean (Logic)

true or false (No quotes)

str

String (Text)

e.g., "Hello Java" (Capital 'S', double quotes)

```
VariablesDemo.java

public class VariablesDemo {
    public static void main(String[] args) {

        // Integer variable
        int age = 18;

        // Decimal variable
        double price = 99.50;

        // Character & Boolean
        char grade = 'A';
        boolean isPassed = true;

    }
}
```

ANATOMY OF A DECLARATION

```
int age = 18 ;
```

BASIC OPERATIONS

Printing & Concatenation

>_ Output to Console

How to Display Data?

In Java, we use `System.out.println()` to send text or numbers to the screen (console).

KEY TECHNIQUES



Printing Text Strings

Text inside double quotes `" "` is printed exactly as written.



Printing Variables

Use the variable name **without quotes** to print its stored value.



Concatenation (+)

The `+` operator "glues" text and variables together into one sentence.

```
public class PrintDemo {
    public static void main(String[] args) {

        // 1. Print simple text
        System.out.println("Hello, World!");

        // 2. Print a variable
        int age = 18;
        System.out.println(age);

        // 3. Mix text & variable (Concatenation)
        System.out.println("My age is " + age);

    }
}
```

ANATOMY OF CONCATENATION

"My age is " + age → My age is 18

VARIABLES IN ACTION

Student Info Example

</> Real-world Data Types

```
1 public class StudentInfo {
2     public static void main(String[] args) {
3         // Basic student details
4         String name = "Rahul"; // Text
5         int year = 1;
6         String branch = "CSE";
7         double cgpa = 8.5;
8         boolean isHosteller = false;
9
10        System.out.println("=== Info ===");
11        System.out.println("Name: " + name);
12        System.out.println("Year: " + year);
13        System.out.println("Branch: " + branch);
14        System.out.println("CGPA: " + cgpa);
15    }
}
```

Multiple Types

1 We can mix different types in one program: `int` for numbers, `double` for decimals, and `String` for text.

String Capitalization

2 Notice the capital 'S' in `String`. In Java, `String` is a class, unlike primitive types like `int` or `boolean`.

Boolean Values

3 Boolean values are `true` or `false`. They are special keywords, so they do not use quotation marks.

LEARNING OUTCOMES

Day 1 Checklist

☰ Goals & Objectives

By the end of today's session, you should be comfortable with these core concepts. **Check yourself against this list:**



Core Understanding

Understand what Java is, why it is platform-independent (JVM), and where it is used in the industry.

1



Program Entry

Know that every Java program starts execution from the `main` method.

2



The Workflow

Be able to write code (.java), compile it (javac), and run it successfully to see output on the console.

3



Variables & Types

Confidently declare and print basic types: `int`, `double`, `boolean`, and `String`.

4



Code Habits

Use single-line (`//`) and multi-line comments to document your logic.

5



Explain Logic

Read the `StudentInfo` example and explain exactly what each line does to a peer.

6